

Past short courses <https://www.ibc2024.org/events/shortcourse24>

<https://www.ibc2024.org/events/ibc2024speakerdetails/short-course-presenter-guide>

IBC 2026 Short Course Proposal

Short Course Title – 125 character max

Motivation – 2500 characters max

- Describe the motivation or the course. Why is it important and timely to present this topic, with reference to current developments in theory and applications? Also, provide an overview of the proposed course content.

Prerequisites: Describe the difficulty level of the course and what prior knowledge/experience is required by the participants.

Outline: Provide a detailed description of what will happen in each course session

Learning Objectives: Please provide a clear and concise statement of learning outcomes. Identify what knowledge, skills, and/or attitudes participants are expected to achieve/demonstrate as a result of the course

* = Required Fields

* **Session Title (Please DO NOT USE ALL CAPITAL LETTERS)**

* **Motivation**

Precision medicine promises to match the right treatment to the right person at the right time. As evidenced by the 2021 special issue of the Journal of the American Statistical Association titled “Theory and Methods Special Issue on Precision Medicine and Individualized Policy Discovery,” precision medicine is one of the most prominent and promising emerging methodological fields in biostatistics. Moreover, the number of ongoing clinical trials focused on developing precision medicine evidence in the context of back pain, behavioral health, and diabetes, among others, continues to grow, with over 200 trials in 2022, a near tenfold increase over the 10 years prior. Motivated by the potential of precision medicine, both methodologically and in clinical research, the goal of this course is to introduce participants to contemporary precision medicine: (1) the identification of heterogeneous treatment effects, (2) the construction of individualized treatment rules, and (3) sequential multiple assignment randomized trials (SMARTs).

This course will emphasize conceptual understanding, methodological rigor, and practical application, with implementation examples (in R) included.

* Prerequisites

We recommend participants to have:

- An understanding of concepts in statistical inference, specifically conditional probability, hypothesis testing, and linear models.
- Familiarity with causal inference concepts such as potential outcomes, confounding, and common assumptions such as the Stable Unit Treatment Value Assumption (SUTVA)
- Basic experience with R programming - loading packages, data handling, and building linear models

Prior exposure to topics such as propensity score methods, machine learning, or clinical trial design is helpful but not required. Though our case studies will focus on certain biomedical or public health areas, we do not require participants to have an understanding of clinical health applications or disease areas.

* Outline

0:00-0:10 Course and instructor introductions

- Welcome and overview of the course objectives
- Introduction of instructors and their areas of expertise
- Brief outline of the course structure and how modules connect

0:10-1:25 Module 1: Heterogeneous treatment effects

- Key Concepts:
 - Average Treatment Effect (ATE), Conditional ATE (CATE), Individual Treatment Effect (ITE)
 - Importance of heterogeneity in treatment responses
- Methodological Approaches:
 - Regression-based modeling
 - Causal forests and nonparametric methods
 - Meta-learners (T-learner, S-learner, X-learner)
- Practical Component:
 - Case study application
 - Hands-on code walkthrough

1:25-2:40 Module 2: Individualized treatment rules

- Key Concepts:
 - Individualized Treatment Rules (ITR), Dynamic Treatment Regimes (DTR), Value function
 - Clinical and policy motivations for ITRs
 - Precision medicine and ITR rules under resource constraints
- Methodological Approaches:

- Augmented Inverse Probability Weighted Estimation (AIPWE)
 - Value-search methods
 - Classification-based approaches (e.g., outcome-weighted learning)
 - Q-learning and constrained Q-learning
- Practical Component:
 - Case study illustrating ITR derivation and real-data application using R package DynTxRegime
 - Applying ITRs in resource-constrained health settings

2:40-3:55 Module 3: Sequential multiple assignment randomized trials (SMARTs)

- Key Concepts:
 - SMART design principles
 - Embedded DTRs, replication, and inverse probability weighting
 - Handling intercurrent events and decision points
- Methodological Approaches:
 - Design considerations and common estimands
 - Estimation strategies tailored to sequential decision-making
 - Practical Component:
- Case study featuring a SMART dataset
 - Coding exercise to construct, evaluate, and compare embedded regimes

3:55-4:00 Closing and discussion

- Summary of key takeaways from each module
- Recommendations on further reading and continued learning resources
- Q&A with instructors

* Learning Objectives

Participants of this short course will be able to:

- (1) Define key precision medicine terminology and understand associated concepts in causal inference, statistical modeling, and machine learning
- (2) Describe the foundations of treatment effect heterogeneity from a causal perspective and explore estimation techniques of causal treatment effect
- (3) Learn to derive and evaluate treatment rules tailored to individual characteristics for decision-making
- (4) Understand the application of precision medicine principles to adaptive interventions, clinical trial strategies, experimental design, and beyond
- (5) Gain hands-on knowledge of working on real-world data using proper precision medicine methods

* Proposed length of course

Half day

Full day

Target Audience

Based on the following scale of 1-5, mark the appropriate score for each category below.

- * Academic statisticians working on the development of statistical methodology (including graduate students)

None selected ▼

- * Epidemiologists/statisticians working in the field of epidemiology

None selected ▼

- * Statisticians working on applications from clinical medicine

None selected ▼

- * Statistical geneticists/statisticians working on genomic data

None selected ▼

- * Statisticians working in the pharmaceutical industry

None selected ▼

- * Statisticians working on applications from ecology/forestry/agriculture

None selected ▼

- * Providers of statistical education: university lecturers, teachers, etc.

None selected ▼

- * Statisticians from developing countries

None selected ▼

- * Statisticians from the local region, the host region for the conference

None selected ▼